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ORIGINAL ARTICLE

Characterization and manufacturing of a paraffin wax as fuel for hybrid rockets

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KEYWORDS

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Abstract The hybrid propulsion performed with paraffin waxes exhibits most attractive capabilities compared to solid or liquid engines, e.g., throttleability and re-ignition, alongside higher regression rates compared to the conventional hydroxyl terminated polybutadiene (HTPB) hybrid fuel. This is because the paraffin wax forms a thin and hydro-dynamically unstable liquid layer, and then enhances the regression rate with the entrainment of droplets from the liquid-gas interface. Nevertheless, some critical open points on the manufacturing of the paraffin fuel grains still persist, because the paraffin wax exhibits high shrinkage during the solidification phase, leading to the formation of cavities, cracks and internal rips, which may be detrimental to the mechanical properties and the structural integrity of the fuel grain. In this context, this paper deals with a wide calorimetric, thermo-mechanical and physical characterization of the paraffin wax selected to manufacture the hybrid rocket engines (HRE) fuel grain, in order to gain a thorough knowledge of the material necessary to avoid the formation of critical defects. Several manufacturing methods were investigated, and it was found that only laboratory scale processes, based on the use of a heated circular mould-piston apparatus, are able to avoid the formation of critical defects, with the application of both high temperature and pressure.

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Nomenclature

a_c	critical size of micro-cracks, voids, etc.
A	burning area
AM	additive manufacturing
DSC	differential scanning calorimetry
EVA	ethylene vinyl acetate copolymer
FDM	fused deposition modeling
FE	finite element
G'	shear storage modulus (unit: Pa)
G''	shear loss modulus (unit: Pa)
h	thickness of the melt layer (unit: m)
HRE	hybrid rocket engines
HTPB	hydroxyl terminated polybutadiene
I	impulse (unit: N•s)
K_{IC}	fracture toughness
$\dot{m}$	mass flow rate (unit: kg/s)
p	dynamic pressure (unit: Pa)
PE	polyethylene
r	average regression rate (unit: m/s)
T	temperature (unit: K)
TGA	thermal-gravimetric analysis
TRL	technology readiness level

Greek symbols

η^*	complex viscosity (unit: Pa•s)
$\dot{\gamma}$	shear rate (unit: Hz)
γ	surface tension in the Owens-Wendt model
μ	melt layer viscosity (unit: Pa•s)
ρ	density (unit: kg/m^3)
σ	surface tension of the melt layer (unit: mN/m)
σ_c	critical stress
θ	contact angle

Subscripts

entr	entrainment
dyn	dynamic
f	fuel
l	liquid
s	solid
sp	specific

1. Introduction

Within the Italian national research program HYPROB-NEW, a specific project is dedicated to hybrid propulsion. The objective is to increase the technology readiness level (TRL) of this system through development and testing of a rocket engine demonstrator of 1 kN thrust class, based on gaseous O_2 and paraffin wax.

Hybrid rockets combine several advantages of solid and liquid rockets engines, overcoming in some cases their disadvantages and limitations. Compared to solid rockets, they are much less sensitive to the propellant cracks and debonds. Compared to a liquid system, the hybrid takes advantage of a reduced explosion hazard and complexity due to the absence of regenerative cooling systems for both the combustion chamber and nozzle. Furthermore, they have high specific impulse I_{sp}, which can be further improved using suitable additives. Moreover, they are throtteable, including shutdown and restart on demand. Additionally, they are safe compared to solid propellant rockets, because the oxidizer is not in contact with the fuel until after it is injected into the combustion chamber. One of the few disadvantages of the hybrid rocket engines (HRE) is the low burning rate, which was overcome with the use of paraffin waxes. They form a thin, hydro-dynamically unstable liquid layer on the fuel-melting surface, thus enhancing the regression rate with the entrainment of droplets from the liquid-gas interface [1–4] (see Figure 1). Regression rates of paraffin waxes, in fact, are 3 to 5 times higher than conventional fuels for HRE, i.e., hydroxyl terminated polybutadiene (HTPB). Compared to the classical HTPB system, paraffin waxes also offer several other advantages, i.e., non-toxic, non-hazardous, shippable as freight cargo, low cost, and reusable (recycling possibilities). In addition, it they possess the same energy per unit mass as kerosene, but their density is 16% greater, no scrap possibility, and a long shelf life [3–6]. Finally, since this fuel is non-explosive, grains can be fabricated on-site and thus can save both manufacturing and launch operation costs [7,8]. In spite of the aforementioned advantages, which encourage the replacement of HTPB, paraffin waxes have two disadvantages: poor mechanical properties and complex manufacturing process.

In order to improve the poor mechanical properties, several authors have formulated blends of paraffin wax with polymers, i.e., polyethylene (PE) [9–12] or ethylene vinyl acetate copolymer (EVA) [9,13,14]. In some cases, aluminum or other energetic additives were added, with the aim of improving the reduced regression rate [10].

Also the manufacturing process of fuel grains remains an open point, especially when grains with large diameters have to be manufactured. In fact, the high shrinkage of about 15%–25% [7] occurring during the solidification of the paraffin wax, leads to grain deformations, internal stresses, rips, defects, micro-cracks and other microstructural discontinuities.

Additionally, the intrinsic brittle behaviour of this material reduces its workability, and finally its thermal inertia hinders a uniform cooling of the whole fuel grain. Obviously, the influence of these phenomena increases as the diameter of the fuel cylindrical grain increases. In addition, during operations, the fuel grain is exposed to a pressure of about 40 bar (for the 1 kN on-ground demonstrator of HYPROB-NEW HRE [15]), so cracks or voids can potentially cause a detachment of large pieces of fuel, which could obstruct the rocket nozzle. For all these

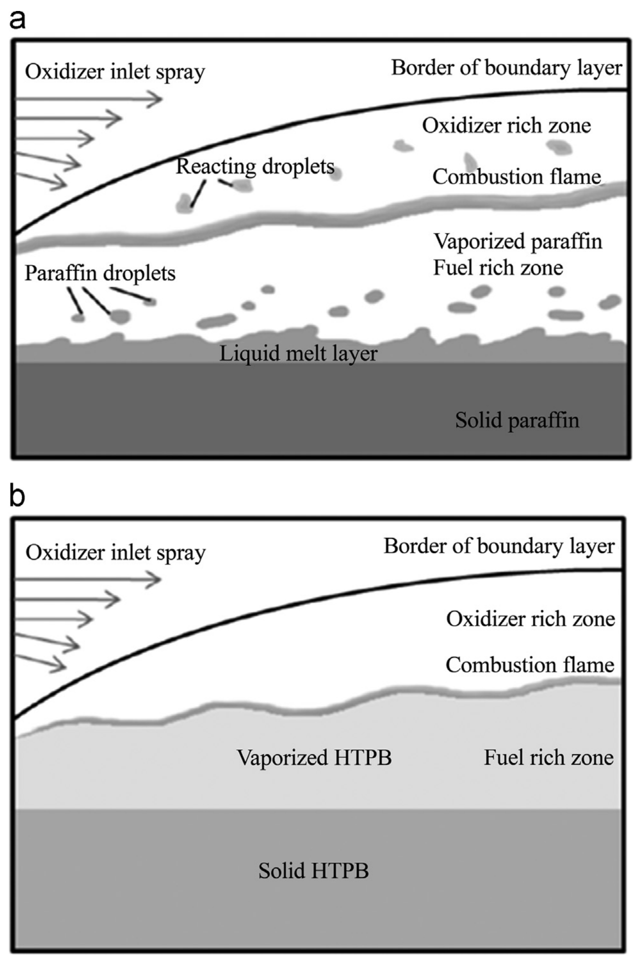

Figure 1 Schematic comparison between (a) liquefying solid fuel (e.g., paraffin wax) and (b) non-liquefying solid fuel (e.g., HTPB).

reasons, the fabrication of a uniform solid fuel grain is of crucial importance. Some authors solved this issue by using a circular mould-piston apparatus able to compensate for the volumetric shrinkage occurring during the paraffin phase transition [16]. According to this method, the formation of voids is minimized by using a piston, since the air trapped during the melting process can be removed.

The effectiveness of a suitable thermal control has been investigated by Gerbal et al. [17] by means of finite element analysis (FEA). In fact, they reduce the volumetric shrinkage by decreasing the temperature gradients in the paraffin grain. De Sain et al. [6] preferred the use of a special spin casting machine with a variable rotational speed, according to which molten wax sealed in a cylindrical container is spun around the desired port axis by a centrifugal force, and is solidified as the system cools down. The spinning rate is conveniently changed in order to distribute the additives homogeneously, e.g., carbon black, along the cross section of the wax fuel grain. In addition, by forcing the liquid paraffin wax to grow on the lateral paraffin wall, the effects of the shrinkage are completely removed. Simultaneously, the centrifugal force forms the central port yielding a glass-smooth finish, with tolerances up to ± 0.7 mm [6].

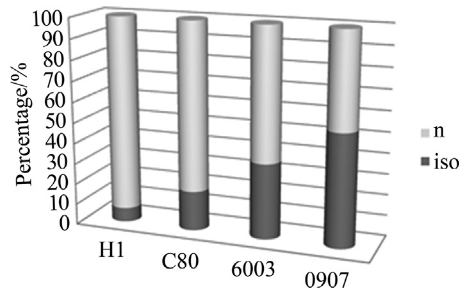

Figure 2 Content of linear (n) and branched (iso) alkanes.

Some researchers experimented using the innovative methods of additive manufacturing (AM), e.g., the layer-by-layer fused deposition modeling (FDM), 3D printing procedure, to manufacture homogeneous solid grains with ports of complex geometry and consequent different thrust profiles having higher consistency, lower costs and shorter lead-times in comparison with the conventional casting methods. Additional benefits of the 3D printing method consist in the ability to produce large grain sections without critical defects and with excellent accuracy (see Stratasys for a 3D world™ technology). Arnold [18] used a machine originally designed to produce solid acrylic parts with paraffin printed as a support to fill voids, to produce paraffin fuel grain structured with honeycomb acrylic features. Theoretically, this method allows the dispersion of energetic additives, but due to lack of previous work on the subject, the authors preferred to subdivide the manufacturing into two steps, i.e., the 3D printing of the tubular honeycomb acrylic structure, and then the spin casting process of molten paraffin wax and additives around the acrylic structure.

The goal of this paper is to find a method to fabricate fuel grains of paraffin wax free of defects, voids and cracks. To this aim, an extensive calorimetric, and physical characterization of the selected paraffin wax, i.e., SASOL® 0907, was performed and discussed here.

Firing test campaigns on intermediate sub-scale motors with 200 N thrust class, are ongoing in order to verify characteristics of the selected paraffin wax (SASOL® 0907) under combustion conditions, using the test bench of Industrial Engineering Department of University of Naples Federico II, located at Grazzanise (Italy).

2. Materials and methods

2.1. Fuel selection

Four different waxes were considered as potential fuels for hybrid rockets, i.e., H1, C80, 6003 and 0907, all manufactured by the SASOL®. They are composed of linear (n), and branched (iso) hydrocarbons, in different percentages, as shown in Figure 2. In addition in Figure 3 the molecular weight distribution of linear and branched alkanes in SASOL® 0907 paraffin wax (data provided by SASOL) is

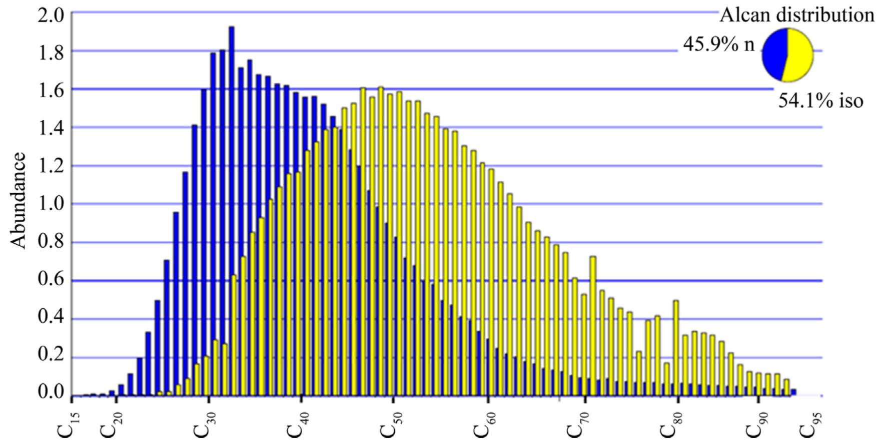

Figure 3 Molecular weight distribution of linear and branched alkanes in SASOL® 0907 paraffin wax.

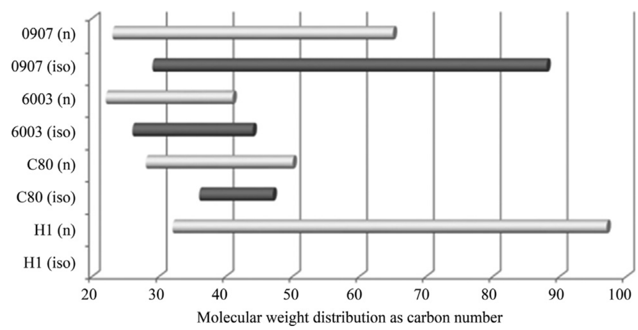

Figure 4 Molecular weight distribution (higher than the 10 wt% vs the maximum abundance) as Carbon number in linear (n) and branched (iso) fractions.

Table 1 Wax properties declared by the manufacturer SASOL.

	Melting or droppling point/°C	Congealing point/°C	Oil content/%	Penetration at 25 °C /(1/10 mm)	Kinematic viscosity at 100 °C /(mm^2/s)	Brookfield viscosity at 135 °C /cP
H1	112	96–100		1		6–8
C80	88	78–83		6	9.4	
6003		60–62	0–0.5	17–20		
0907	88–102	83–94	0–1.5	4–10	14–18	

reported. Moreover, the distribution of their molecular weights is centred on different values, as shown in Figure 4, where percentages higher than the 10% by weight of the maximum abundance is reported as a function of the carbon number, for both linear and branched fractions.

Table 1 shows the wax properties cited by the manufacturer SASOL® in terms of melting and congealing points, penetration and viscosity. It appears that the H1 formulation consists predominantly of linear alkanes having a large distribution of the molecular weights, i.e., ranging between C_{32} and C_{97} centered at C_{65}. SASOL® 6003 has an abundance of linear alkane chains with respect to the branched alkanes, but both linear and branched molecular weight distributions are centered on the same value. Unlike the other paraffins, the formulation SASOL® 0907 is predominantly composed of branched alkanes, whose content is also the highest. Additionally, the molecular weight distribution of the SASOL® 0907 branched alkanes is centred on the highest value.

The high fraction of branched iso-alkanes in SASOL® 0907 may be more advantageous to form microcrystals, which allows higher fracture toughness and thus a better workability during the manufacturing of the fuel grain in comparison with the linear paraffin waxes. Moreover, paraffin solid wax is an intrinsically fragile material at least at the investigated operating conditions. For this reason and according to Griffith theory, higher values of fracture toughness increase the tolerance of surface or internal micro-cracks and other microstructural defects of increased radius, keeping constant, at the same time the applied thermo-mechanical stress level during operations.

For all these reasons, the SASOL® 0907 was selected to manufacture the fuel grains for the HYPROB-NEW HRE technology demonstrator.

2.2. *Differential scanning calorimetry (DSC)*

Differential scanning calorimetry (DSC) measurements were performed using a TA instruments DSC Q20. Every test was carried out by firstly cooling the paraffin wax down to 0 ℃, to have a suitable baseline, and then heating it up to 120 ℃ with heating rates of 2, 10, and 40 ℃/min. A possible modification of the paraffin crystal structure, caused by the cooling step during the manufacturing process, could affect its integrity and consequently the mechanical performance of the fuel grain. Thus, to study the effect of the cooling rate on the paraffin crystallization, also the cooling steps at 2, 10, and 40 ℃/min and the re-heating at 10 ℃/min of the SASOL® 0907 paraffin wax were monitored.

2.3. *Thermo-gravimetric analyses (TGA)*

Thermo-gravimetric analyses (TGA) were performed using the Discovery TGA by TA Instruments. The pellets of paraffin wax were heated from 40 ℃ to 700 ℃ at 10 ℃/min, using both nitrogen and air controlled atmospheres.

2.4. *Surface tension measurements*

The surface free energy was calculated according to the Owens-Wendt method [19]. The Owens-Wendt approach is one of the most commonly used methods to calculate the surface free energy of materials [20]. The main assumption of this method is that the surface free energy is the sum of two components: the dispersion and the polar components [21,22]. The Owens-Wendt model is described by the geometric mean relationship:

$$\frac{1}{2}(1+\cos\theta)\gamma_L = \left(\gamma_S^D\cdot\gamma_l^D\right)^{\frac{1}{2}} + \left(\gamma_S^P\cdot\gamma_l^P\right)^{\frac{1}{2}} \tag{1}$$

where θ is the contact angle between the liquid droplet and surface, γ_L the liquid total surface tension, γ_l^D the dispersion component of liquid surface tension, and γ_l^P the polar component of liquid surface tension. The unknown terms in the equation are γ_S^D, which is the dispersion component of the solid surface free energy, and γ_S^P, namely the polar component of the solid surface free energy.

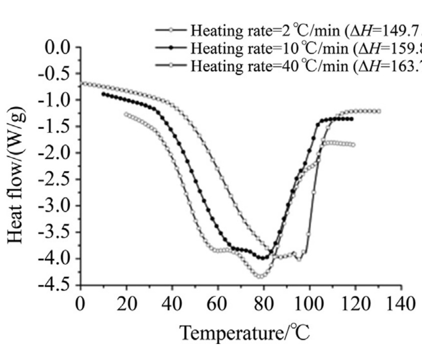

Figure 5 DSC thermograms at different heating rates (endothermic down).

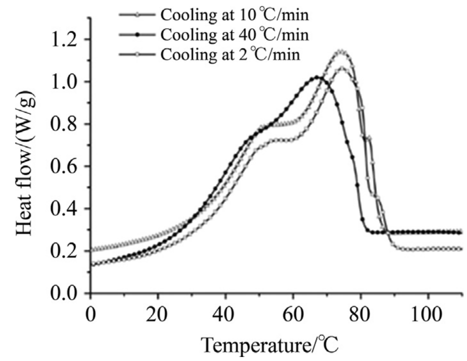

Figure 6 DSC thermograms at different cooling rates (endothermic down).

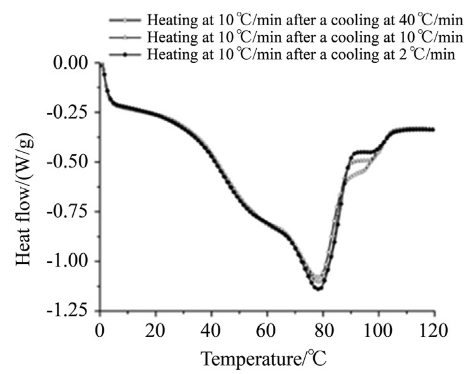

Figure 7 DSC thermograms at 10 ℃/min of samples cooled at different cooling rates (endothermic down).

Table 2 Parameters obtained from DSC measurements.

Heating/cooling rate/ (℃/min)	$T_{m,1}$/ ℃	$T_{m,2}$/ ℃	$T_{m,3}$/ ℃	$T_{c,1}$/ ℃	$T_{c,2}$/ ℃	$T_{c,3}$/ ℃
2	60	78	102.5	54	74	85
10	67	81	97	54	74	83
40	87	95.5		51	67	77

The value of the total surface free energy of the solid is obtained as $\gamma_S=\gamma_S^D+\gamma_S^P$. Dividing by $\left(\gamma_l^P\right)^{\frac{1}{2}}$, the Eq. (1) can be re-written as:

$$\frac{\frac{1}{2}(1+\cos\theta)\gamma_L}{\left(\gamma_l^P\right)^{\frac{1}{2}}} = \left(\frac{\gamma_l^D}{\gamma_l^P}\right)^{\frac{1}{2}}\cdot\left(\gamma_S^D\right)^{\frac{1}{2}} + \left(\gamma_S^P\right)^{\frac{1}{2}} \tag{2}$$

The left-hand side of Eq. (2) contains quantities which are measured experimentally (θ) or which are available in the literature (γ_l^P, γ_L), so that a plot of the left-hand side of Eq. (2) versus quantity $\left(\frac{\gamma_l^D}{\gamma_l^P}\right)^{\frac{1}{2}}$ gives a straight line with slope $\left(\gamma_S^D\right)^{\frac{1}{2}}$ and intercept $\left(\gamma_S^P\right)^{\frac{1}{2}}$.

To assess the surface free energy of SASOL® 0907 paraffin wax, measurements of contact angle using liquids with different polarities, i.e., water, formamide, and

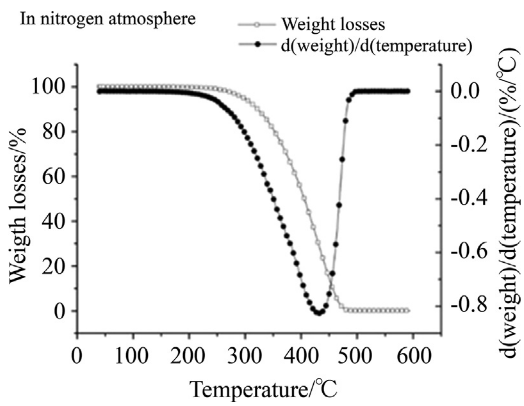

Figure 8 Thermogravimetric analysis in nitrogen atmosphere.

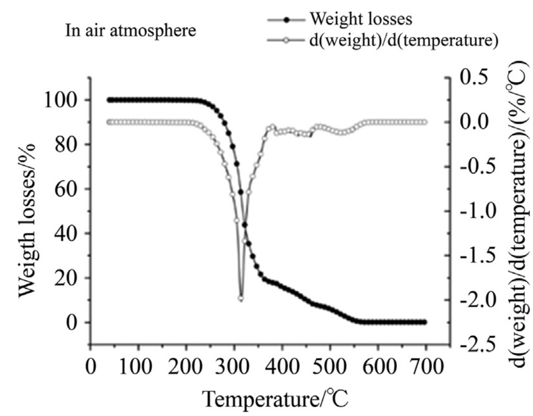

Figure 9 Thermogravimetric analysis in air atmosphere.

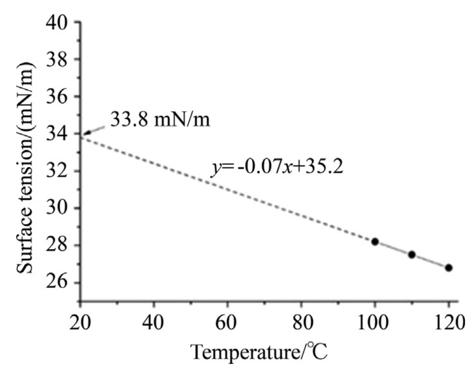

Figure 10 Surface tension as a function of the temperature.

diiodomethane, were performed. For each liquid the contact angle values are averaged over ten measurements.

2.5. *Rheological characterization*

The rheological characterization was performed using a TA Instruments HR-2 Hybrid Rheometer with 40 mm parallel plate geometry and 500 µm gap distance. Measurements of storage and loss moduli, and complex viscosity, in temperature sweep mode were performed on both solid and melted paraffin waxes. In detail, a heating rate of 2 ℃/min was set for the solid state between 30 and 80 ℃, and 20 ℃/min for the temperature sweep from 100 to 200 ℃. A frequency of 1 Hz was set for all tests. In addition, a frequency sweep ranging from 0.1 to 100 Hz, i.e., from 0.6 to 628.5 rad/s, at 150 ℃ was carried out. This was because the melt layer viscosity has a great influence on the regression rate [1], since, as previously reported [20], it affects the entrainment contribution of the mass flow rate, $\dot{m}_{entr} \propto \frac{p_{dyn}^{\alpha} h^{\beta}}{\mu_l^{\gamma} \sigma^{\pi}}$ where, according to Karabeyouglu et al. [1,2]. p_{dyn} is the dynamic pressure, h the thickness of the melt layer, σ the surface tension and μ_l the melt layer dynamic viscosity. Exponents values α, β, γ, π can be found in the literature from Gater and L'Ecuyer [23], Nigmatulin et al. [24], and Karabeyouglu [1].

2.6. *Fractographic analysis*

Fracture surfaces of solid grain paraffin wax samples, manufactured according to various methods, were inspected by means of a stereo microscope the LEICA MZ-12X whereas the micro-fractography was conducted using an optical microscope the LEICA DM-RXE in bright field and using an optical magnification of 550x.

3. Experimental results

3.1. *Differential scanning calorimetry (DSC)*

The DSC results at different heating rates of paraffin pellets are shown in Figure 5 (endothermic down). Two different peaks and a shoulder at the highest temperatures are apparent, as evident at the lowest heating rate, i.e., 2 ℃/min. The affinity between the shape of thermograms and the molecular weight distribution, shown in Figure 3, suggests that there may be a relationship between them. Hence, by analyzing the DSC thermogram obtained at 2 ℃/min, the peak at 60 ℃ could correspond to the melting of the molecular weight fraction centred at about C_{37}, whereas the peak at 78 ℃ could correspond to the melting of the molecular weight fraction centred at about C_{50}, and finally, the peak at 102.5 ℃ could be related to the melting of the molecular weight fraction centred at C_{83}. Differences in sharpness of peaks among thermograms at different heating rates can be attributed to the thermal inertia of the paraffin wax. Figure 6 shows the thermograms corresponding to the different cooling rates of the molten paraffin. Again, no relevant differences can be observed apart from the delay in temperature and the sharpness of peaks.

Figure 7 shows the thermograms related to the heating scans at 10 ℃/min after three different cooling rates. Some differences appear only at 90–100 ℃, temperatures at which the highest molecular weights of branched alkanes are able to melt, as the molecular weight distribution of Figure 3 suggests. Probably, this fraction is able to organize itself into different crystalline structures when it undergoes different cooling rates.

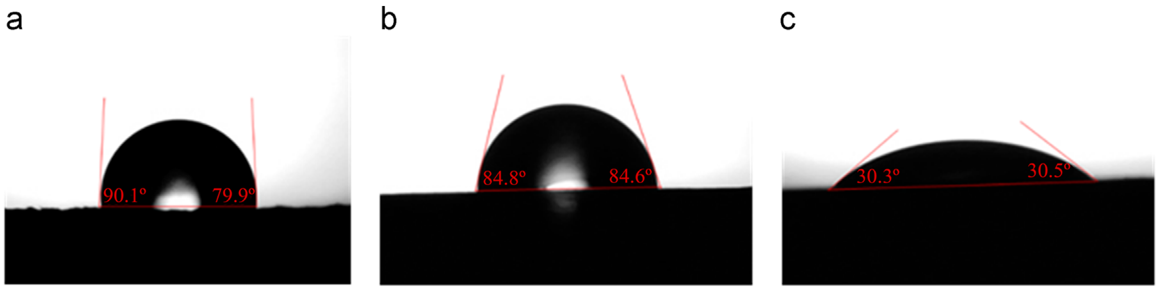

Figure 11 The three different shapes of the liquid droplets ((a) distilled water, (b) formamide, and (c) diiodomethane) in contact with the paraffin sample.

Table 3 Surface tension parameters.

Wettability	[°]	87±7
Surface free energy	[mJ/m^2]	35
Dispersion component	[mJ/m^2]	33
Polar component	[mJ/m^2]	1
Work of adhesion paraffin-water	[mJ/m^2]	77

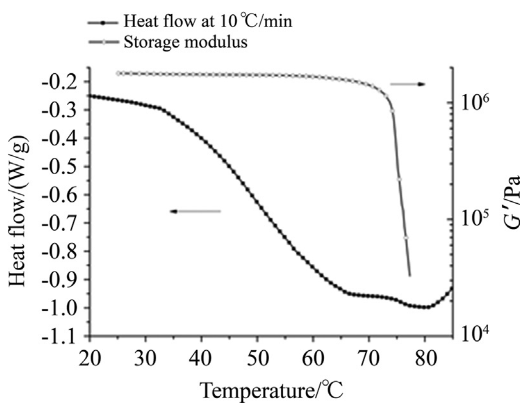

Figure 12 Comparison between storage modulus at 1 Hz and heat flow at 10 ℃/min as a function of the temperature.

In Table 2 all data provided by the DSC measurements are summarized. They differ slightly from those reported in the literature [25]. In fact, at a heating rate of 10 ℃/min a value of $T_{m,1}$ equal to 67 ℃ instead of 59.7 ℃ was measured. Similar differences were observed for $T_{m,2}$ equal to 81 ℃ instead of 82.1 ℃, and a $T_{m,3}$ equal to 97 ℃ instead of 98.9 ℃. Slight differences appear also during the cooling ramp, particularly a value of 54 ℃ instead of 47.1 ℃ for the $T_{c,1}$, was measured and similarly a value of 74 instead of 70.0 ℃ for the $T_{c,2}$, and a value of 83 instead of 83.8 ℃ for the $T_{c,3}$. These differences can presumably be attributed to the different instruments used.

In general, the wide melting and solidification ranges observed are probably due to the high percentage of branched alkanes, which gradually undergo the phase transition [25].

To summarize, regarding the manufacturing of the fuel grain, these analyses suggest that at 110 ℃ and at atmospheric pressure the paraffin wax pellets are completely liquefied, independently of the heating rate. The cooling step of the manufacturing process of the fuel grain has no relevant effects on its final crystal structure, since it affects only the highest branched molecular weight fraction, which represents only a small percentage of the total.

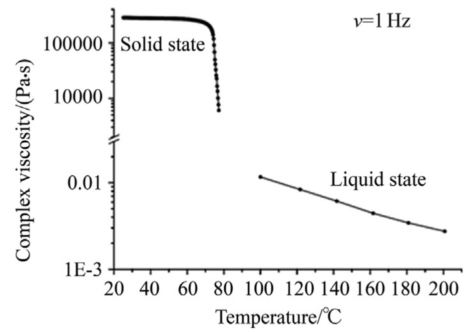

Figure 13 Dynamic viscosity of paraffin pellets in solid and liquid states.

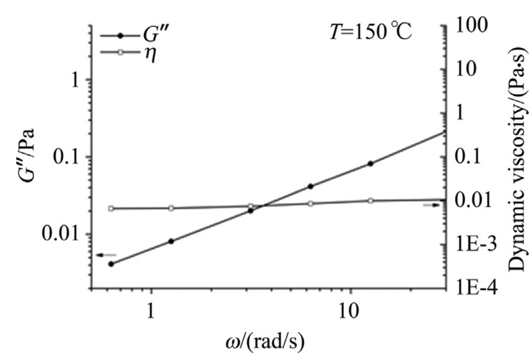

Figure 14 G'' and viscosity measured in frequency sweep mode at 150 ℃.

3.2. Thermo-gravimetric analyses (TGA)

The thermogravimetric results illustrated in Figures 8 and 9 highlight that the weight losses at temperatures lower than 110 ℃, the maximum temperature at which the investigated paraffin wax is subjected, during the fuel grain manufacture, are negligible. In fact, only at 190 ℃ the weight losses

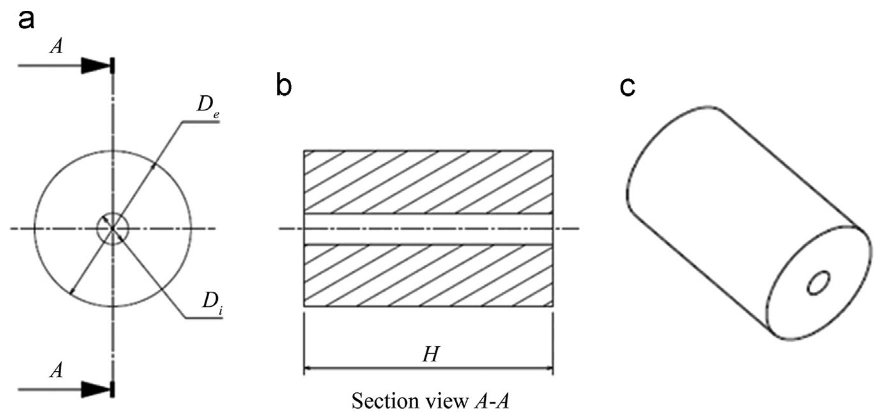

Figure 15 CAD of paraffin wax solid fuel grain sample configuration. (a) Cross section, (b) longitudinal section, and (c) whole shape of the paraffin grain.

Table 4 Size of solid paraffin wax grain configurations.

Configuration	Height H/mm	External diameter D_e/mm	Internal diameter D_i/mm	Aspect ratio
1	80	65	Not present	0.8
2	95	50	Not present	0.5
3	35	50	4	1.4

become significant reaching the value of 0.1 wt%. This is evidence that volatile elements are almost absent at temperatures involved in the production process of the fuel grain, so the process should be considered completely safe. The thermo-gravimetric analysis performed in a nitrogen atmosphere (see Figure 8) shows a narrow and well-defined thermal event, which corresponds to the boiling point of the paraffin wax occurring at about 430 ℃. In addition, the analysis performed in an air atmosphere (Figure 9) highlights that, after the main oxidative degradation at about 300 ℃, there is a significant residual, ascribed to the formation of ashes, which are difficult to oxidize further at temperatures below 550 ℃. The formation of these ashes could affect the re-ignition process of the fuel grain during the firing test.

3.3. Surface tension measurements

The surface free energy was measured on solid paraffin pellets of SASOL® 0907, by putting them in a silicone mold and melting then at 110 ℃. The paraffin wax sample was solidified very slowly in order to avoid the formation of possible macro and micro-cracks due to the liquid-solid transition [26]. Through this procedure, it is possible to realize a paraffin sample with a very smooth surface, where the measured surface free energy will depend only on the physical state of the paraffin, and not on its morphology. In fact, during the melting process, it is evident that a very smooth layer of melted paraffin pellets is obtained due to the surface tension of the liquid paraffin. It is worth noting, that the surface tension is correlated to the liquid paraffin wax, whereas the surface free energy is related to the solid paraffin wax. The surface free energy is expressed in [J/m^2], which is equal to [Nm/m^2], whereas the surface tension is measured in [N/m], consequently the two performance indexes can be compared. In order to obtain the surface tension value at high temperature, i.e., corresponding to the liquid state of SASOL® 0907 paraffin wax, a measure at 20 ℃ of the surface tension of the solid paraffin wax was carried out, giving in output a value of 34.78 mN/m. This value is in good agreement with the corresponding value of 33.8 mN/m obtained by linearly extrapolating (see Figure 10) the surface tension values measured by Kobald et al. [26,27], meaning that the value of 27.5 mN/m at 110 ℃ can be used.

The results in Figure 11 show the shape of liquid droplets on the paraffin wax surface. All results were processed and the performance indexes are summarized in Table 3.

It is worth noting that water droplet on the SASOL® 0907 paraffin wax surface describes a partially hydrophobic behavior. This was confirmed also by analyzing the contact angle measurements on formamide, partially polar liquid with respect to the water. Whereas the paraffin wax highlights a hydrophilic behavior with diiodomethane due to its apolar nature.

3.4. Rheological and viscosity measurements

The storage modulus and complex viscosity of paraffin wax in the solid state are shown in Figure 12 and Figure 13, respectively. In particular, Figure 12 shows the comparison between the heat flow during the DSC heating scan at 10 ℃/min and the storage modulus, highlighting that paraffin wax has a storage modulus of 1.7 × 10^6 Pa, in the solid state, up to the first melting peak at about 70 ℃,

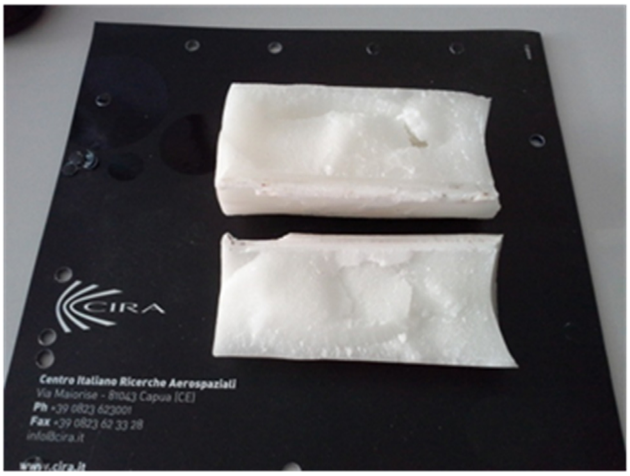

Figure 16 Paraffin wax solid grain sample manufactured with the laboratory method at 80 °C as peak temperature.

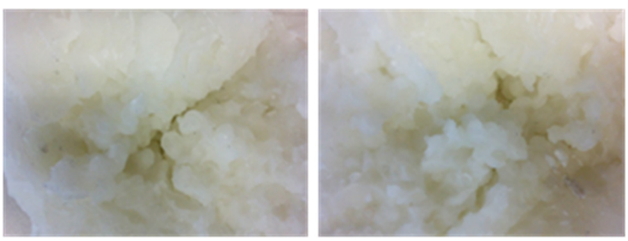

Figure 17 Fracture surfaces of paraffin wax solid grains manufactured with the laboratory method at 80 °C as peak temperature.

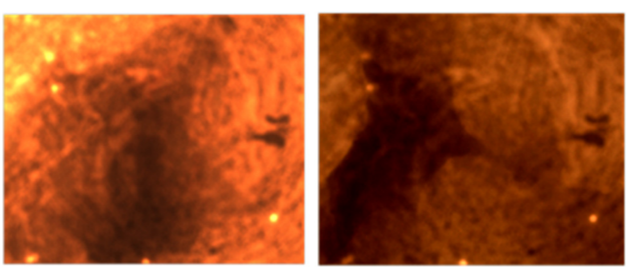

Figure 18 Optical microscope images of paraffin wax solid grain samples manufactured with the laboratory method at 80 °C as peak temperature.

temperature at which the SASOL® 0907 paraffin wax ceases to preserve its elastic response anymore. Similarly, the complex viscosity (Figure 13) ranges between 2.8 and 2.2×10^5 Pa·s before melting, whereas in the liquid phase, the viscosity decreases as the temperature increases, as expected, with an Arrhenius-type behavior. In Figure 14 the loss modulus G" and dynamic viscosity measured in frequency sweep mode at 150 °C are illustrated. It should be noticed that in the frequency range from 0.6 to 30 rad/s, the melted paraffin wax displays a Newtonian behavior.

It is concluded that, the viscosity does not change with angular frequency ω, whereas the loss modulus G" increases linearly as the angular frequency ω increases.

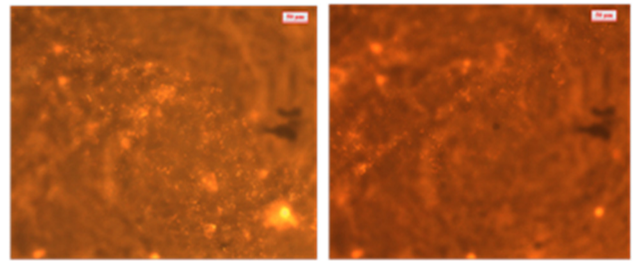

Figure 19 Optical microscope images obtained in bright field using a magnification of 550x of paraffin solid grains manufactured with the laboratory method at 80 °C as peak temperature.

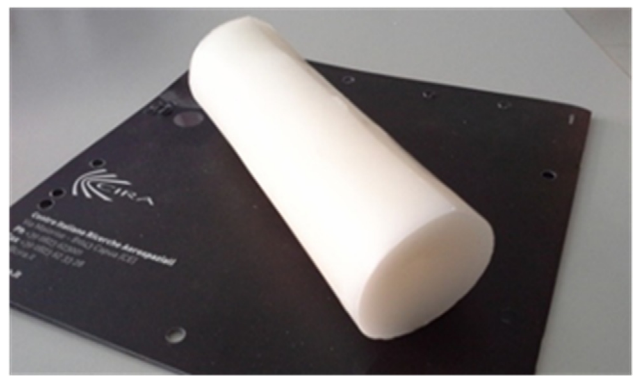

Figure 20 Paraffin wax solid grain sample manufactured with the laboratory method at 110 °C as peak temperature.

In the $\dot{m}_{entr}$ formula a value of viscosity at an average temperature, ranging between the melting and boiling (about 450 °C) points, calculated using equation from Marano and Holder [28,29] must be introduced. Since it is not possible to perform measurements at same high temperatures, the value of viscosity must be extrapolated starting from values of complex viscosity obtained in the range 100–200 °C, using the asymptotic behavior correlations (ABC) attributed to Kreglewski and Zwolinski [30]. This is a method which predicts PVT, thermal and transport properties of n-paraffins and n-olefins; and it was also demonstrated that it could be successfully applied to branched alkanes if the length of branched chains are limited [30]. The Newtonian behavior of the liquid SASOL® 0907 paraffin wax suggests that the branched length chains are short enough to not affect the rheological outputs where in this case, the viscosity errors are limited to 8% [30]. According to the ABC method, the viscosity can be assessed using the following equations:

$$Y = Y_{\infty,0} + \Delta Y_{\infty}(n - n_0) - \Delta Y_0 \exp(-\beta(n - n_0)^{\gamma}) \quad (3)$$

$$\Delta Y = A + B/T + C\ln T + DT^2 + E/T^2 \quad (4)$$

where Y is the property value, e.g., viscosity, at a carbon number n, Y_0 is the property pseudo-value at $n = 0$, Y_{∞} is the limiting property value as n becomes very high, β and γ are adjustable parameters, A, B, C, D, and E, are temperature dependent ABC parameters, listed in Ref. [30]. The

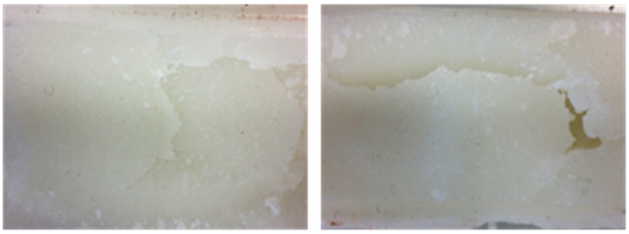

Figure 21 Fracture surfaces of paraffin wax solid grains manufactured with the laboratory method at 110 ℃ as peak temperature.

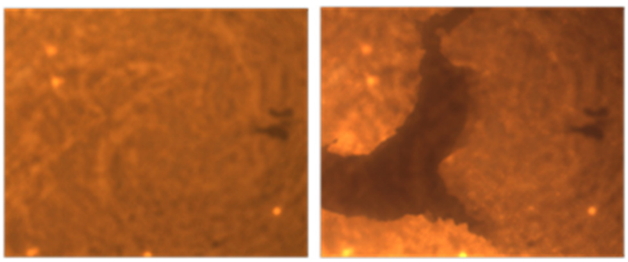

Figure 22 Optical microscope images of paraffin wax solid grain samples manufactured with the laboratory method at 110 ℃ as peak temperature.

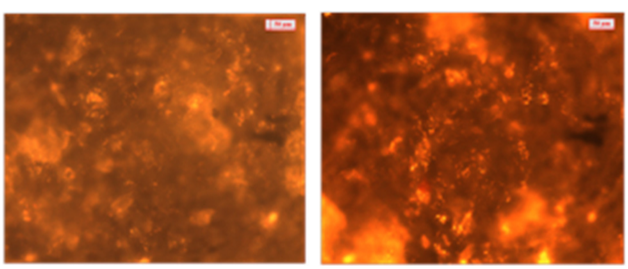

Figure 23 Optical microscope images obtained in bright field using a magnification of 550x of paraffin wax solid grain samples manufactured with the laboratory method at 110 ℃ as peak temperature.

viscosity was assessed at the temperature of 280 ℃, assumed as the initial temperature of the combustion chamber, which is the average temperature between the melting and the boiling (i.e., about 450 ℃) points of the paraffin wax. In particular, a melting point equal to 110 ℃ (see Figure 5) was considered to ensure that all crystals were molten. As reference temperature, the highest available experimental value, i.e., 200 ℃ was used. The value of viscosity at 280 ℃, calculated using the ABC method, is equal to 0.3747×10^{-3} Pa · s.

3.5. *Grain manufacturing*

Several configurations of solid paraffin wax grains were fabricated having a cylindrical shape with a coaxial central port, as indicated in the technical drawing shown in Figure 15.

Samples of different sizes and aspect ratios were manufactured. The height, external diameter and internal port diameter of every configuration are listed in Table 4. For the

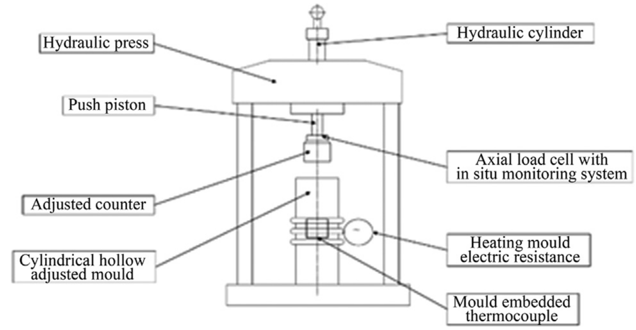

Figure 24 Scheme of in-house device used for paraffin wax fuel solid grains manufacturing.

Figure 25 Paraffin wax solid grain manufactured with the die casting method.

third configuration, the central co-axial port was realized, after solidification, by means of mechanical machining.

In order to highlight the presence of voids and cracks, the paraffin fuel grains, once divided into two parts, were inspected and fractographically analyzed.

Initially, basic laboratory procedures were used for the first preliminary manufacturing tests of the solid fuel grains. This method consisted in weighing the required amount of paraffin wax pellets, inserting the material into a suitable mould, heating to a predetermined temperature, stirring the molten liquid and finally slowly cooling it down to room temperature.

The first experiments were performed using an aluminum cylindrical vessel of the size and shape corresponding to configuration 1 in Table 4. Pellets were heated to 80 ℃ using an electric oven under high vacuum conditions, in order to eventually de-gas any entrapped air. This temperature was selected by the DSC scan (see Figure 5) corresponding to the temperature at which the majority of hydrocarbon chains melt. The liquid paraffin wax was slowly cooled down at about 1 ℃/min to room temperature. The low cooling rate should have reduced the formation of internal and surface cracks, and voids during solidification.

In this first experiment, no external device for applying a mechanical pressure on the molten wax during transition

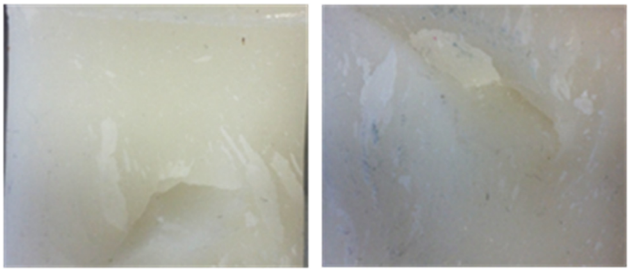

Figure 26 Fracture surfaces of paraffin wax solid grain samples manufactured with the die casting method.

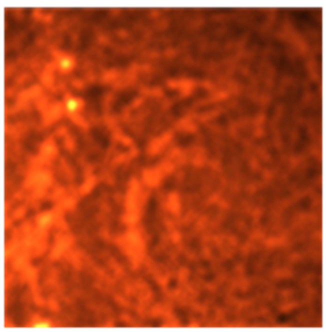

Figure 27 Optical microscope image of paraffin wax solid grain samples manufactured with the die casting method.

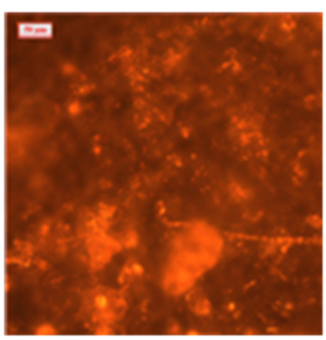

Figure 28 Optical microscope image obtained in bright field using a magnification of 550x of paraffin wax solid grain samples manufactured with the die casting method.

from liquid to solid state was used and a very slow cooling rate was the only method employed to decrease the thermal gradients inside the bulk material. These cylindrical samples were separated along the longitudinal axis and the resulting fracture surfaces were microscopically analyzed in order to investigate the possible presence of internal micro-cracks and micro-voids.

Figure 16 illustrates an image of fracture surfaces of one sample manufactured according to this method, whereas Figures 17, 18, and 19 illustrate the most important features revealed by microscopic analysis. It highlights the edges are homogeneous, but the core region shows traces of pellets, which were only partially liquefied, and large and numerous internal micro-voids and micro-cracks.

From these experiments it is possible to deduce, that 80 °C is not sufficiently high to realize homogenous fuel grains. As shown in Figures 3 and 5, 80 °C corresponds to the temperature at which the majority of linear paraffin chains melt. This result suggests that higher temperatures are needed in order to partially melt also the branched fractions of paraffin. On the other hand, the presence of internal flaws can be attributed to the spontaneous, volumetric shrinkage occurring during cooling, and transition from liquid to solid state.

To improve the quality of the manufactured solid grains, further tests were carried out using a very similar method, but heating the pellets of SASOL® 0907 up to 110 °C inside a polyamide mould, of the size and shape corresponding to the configuration 2 in Table 4. This temperature was chosen because, as suggested by DSC analysis illustrated in Figure 5, this is the temperature at which all the paraffin wax crystals melt. A polyamide mould was selected to increase both thermal insulation and detaching capacity of the solid grains. Photographs of the paraffin wax solid grain obtained in this way are shown in Figure 20. The results of the microscopic analysis are shown in Figure 21, Figure 22, and in Figure 23.

The photographs demonstrate that the temperature of 110 °C is adequate to melt all crystals, but an internal, large area of macro-porosity was detected, again attributed to the spontaneous volumetric shrinkage. Although the very slow cooling was aided by the polyamide mould, it was not sufficient to manufacture homogeneous grains. Indeed, the shrinkage needs to be adequately compensated by an externally applied pressure.

To improve the relatively poor quality of paraffin wax solid grains manufactured with the laboratory method, a second, alternative, more sophisticated procedure was investigated and experimented, based on the simultaneous application of thermal and mechanical energies, i.e., heat flux and pressure.

The starting point was the raw material in the form of commercially available pellets of SASOL® 0907 paraffin wax. The pellets were introduced inside a cylindrical mould, equipped with an embedded electrical resistance heating system and temperature measurements through a K type thermocouple. The paraffin wax was heated at 110 °C for 20 min until it was completely melted. The molten wax was stirred, then the electrical resistance heating system was switched off and simultaneously a pressure of about 10 bar was applied by means of a cylindrical piston, fitted with a load cell for monitoring the exerted compressive force. Finally, the material was slowly cooled down to room temperature.

This manufacturing apparatus was accomplished by the use of specially designed equipment, which is illustrated in Figure 24. A load of 300 kg was set. Considering the surface area of the melted paraffin wax, this value is equivalent to a pressure of about 10 bar, kept constant by means of the hydraulic system. The tight coupling tolerances among the mould components allowed the expulsion of air entrapped inside the paraffin wax during the heating and stirring steps.

Using this method homogeneous, consistent, solid paraffin grains corresponding to the configuration 3 in Table 4 were fabricated (Figure 25). In addition, in this case, the internal microstructure of the samples were microscopically analyzed and the resulting images are shown in Figure 26, Figure 27 and in Figure 28.

It is worth noting that using this method traces of pellets which were not completely liquefied, and surface and internal micro-cracks, and micro-porosities of critical size were not detected.

The lesson learned from all the experiments reported in the present investigation is that simultaneous application of external mechanical pressure of about 10 bar and of thermal loads corresponding to a bulk temperature of at least 110 ℃, are necessary to avoid the abovementioned manufacturing problems, i.e., incomplete melting of the pellets and the presence of micro-porosities.

4. Conclusions

A commercial paraffin wax, i.e., the SASOL® 0907, possessing a high amount of branched alkanes, was selected for its capacity to form micro-crystals, which allow high fracture toughness and workability. A wide thermal and rheological characterization was carried out supporting the manufacturing process. The thermal analyses, i.e., DSC and TGA, gave information about the phase transitions and thermal and thermo-oxidative stabilities. The results show that at atmospheric pressure and 110 ℃ the paraffin pellets are completely melted, and that the thermal treatment used to manufacture the fuel grain once melted essentially does not influence the crystals structure. Additionally, the formation of ashes during the thermo-oxidative degradation of paraffin wax has to be taken into account for the re-ignition tests of the fuel grain. Rheological characterizations gave information on the storage and loss moduli, and viscosity of both solid and liquid paraffin wax. In particular, the viscosity value at high temperature, i.e., 280 ℃, was assessed using an extrapolation method, in order to establish the value to be introduced in the mass flow rate equation. The wide characterization was useful to address the satisfactory manufacture of the fuel grain. Several manufacturing methods were investigated and it was found that the use of a heated circular mould-piston system hinders the formation of voids and critical defects. This tool consists of a cylindrical metallic box, heated through electrical resistance and applying pressure via an oleo-dynamic piston as the paraffin pellets melt. Preliminary firing tests demonstrated the effectiveness of this manufacturing methodology. The next step of the authors will be the scaling-up of the fuel grain sizes and eventually the addition of nanofillers.

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